

(b) Using the formula $I_R = -100/R^2$ from part (a), we find the instantaneous rate of change of I with respect to R at $R = 20$ to be

$$I_{20} = \frac{100}{20^2} = \frac{1}{4}$$

Thus, when $R = 20$, the current is *decreasing* at a rate of $\frac{1}{4}$ ampere per ohm.

EXERCISES 3.1

Exer. 1–6: (a) Use Definition (3.1) to find the slope of the tangent line to the graph of f at $P(a, f(a))$. (b) Find an equation of the tangent line at $P(2, f(2))$.

1 $f(x) = 5x^2 - 4x$

2 $f(x) = 3 - 2x^2$

3 $f(x) = x^3$

4 $f(x) = x^4$

5 $f(x) = 3x + 2$

6 $f(x) = 4 - 2x$

Exer. 7–10: (a) Use Definition (3.1) to find the slope of the tangent line to the graph of the equation at the point with x -coordinate a . (b) Find an equation of the tangent line at P . (c) Sketch the graph and the tangent line at P .

7 $y = \sqrt{x}$; $P(4, 2)$

8 $y = \sqrt[3]{x}$; $P(-8, -2)$

9 $y = 1/x$; $P(2, \frac{1}{2})$

10 $y = 1/x^2$; $P(2, \frac{1}{4})$

Exer. 11–12: (a) Sketch the graph of the equation and the tangent lines at the points with x -coordinates -2 , -1 , 1 , and 2 . (b) Find the point on the graph at which the slope of the tangent line is the given number m .

11 $y = x^2$; $m = 6$

12 $y = x^3$; $m = 9$

Exer. 13–14: The position function s of a point P moving on a coordinate line l is given, with t in seconds and $s(t)$ in centimeters. (a) Find the average velocity of P in the following time intervals: $[1, 1.2]$, $[1, 1.1]$, and $[1, 1.01]$. (b) Find the velocity of P at $t = 1$.

13 $s(t) = 4t^2 + 3t$

14 $s(t) = 2t - 3t^2$

15 A balloonist drops a sandbag from a balloon 160 feet above the ground. After t seconds, the sandbag is $160 - 16t^2$ feet above the ground.

(a) Find the velocity of the sandbag at $t = 1$.

(b) With what velocity does the sandbag strike the ground?

16 A projectile is fired directly upward from the ground with an initial velocity of 112 ft/sec. Its distance above the ground after t seconds is $112t - 16t^2$ feet.

(a) Find the velocity of the projectile at $t = 2$, $t = 3$, and $t = 4$.

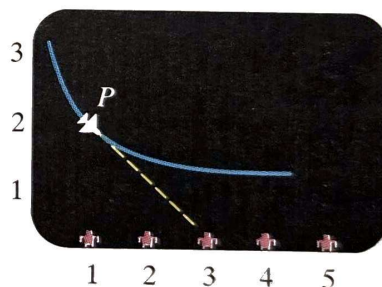
(b) When does the projectile strike the ground?

(c) Find the velocity at the instant it strikes the ground.

17 In the video game shown in the figure, airplanes fly from left to right along the path $y = 1 + (1/x)$ and can shoot their bullets in the tangent direction at creatures placed along the x -axis at $x = 1, 2, 3, 4$, and 5 . Determine whether a creature will be hit if the player shoots when the plane is at

(a) $P(1, 2)$ (b) $Q(\frac{3}{2}, \frac{5}{3})$

EXERCISE 17

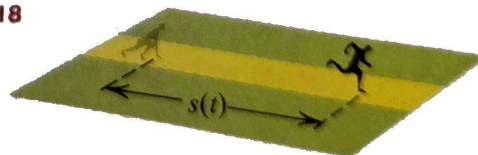


18 An athlete runs the hundred-meter dash in such a way that the distance $s(t)$ run after t seconds is given by $s(t) = \frac{1}{5}t^2 + 8t$ m (see figure). Find the athlete's velocity at

(a) the start of the dash (b) $t = 5$ sec

(c) the finish line

EXERCISE 18



Exer. 19–20: (a) Find the average rate of change of y with respect to x on the given interval. (b) Find the instantaneous rate of change of y with respect to x at the left endpoint of the interval.

19 $y = x^2 + 2$; $[3, 3.5]$

20 $y = 3 - 2x^2$; $[2, 2.4]$

- 21 Boyle's law states that if the temperature remains constant, the pressure p and volume v of a confined gas are related by $p = c/v$ for some constant c . If, for a certain gas, $c = 200$ and v is increasing, find the instantaneous rate of change of p with respect to v at
- (a) any volume v (b) a volume of 10
- 22 A spherical balloon is being inflated. Find the instantaneous rate of change of the surface area S of the balloon with respect to the radius r at
- (a) any value of r (b) $r = 3$ ft
- 23 Graph $f(x) = \sin \pi x$ on the interval $[0, 2]$.
- (a) Use the graph to estimate the slope of the tangent line at $P(1.4, f(1.4))$.
- (b) Use Definition (3.1) with $h = \pm 0.0001$ to approximate the slope in (a).
- 24 Graph $f(x) = \frac{10 \cos x}{x^2 + 4}$ on the interval $[-2, 2]$.
- (a) Use the graph to estimate the slope of the tangent line at $P(-0.5, f(-0.5))$.
- (b) Use Definition (3.1) with $h = \pm 0.0001$ to approximate the slope in (a).
- (c) 25 An object's position on a coordinate line is given by
- $$s(t) = \frac{\cos^2 t + t^2 \sin t}{t^2 + 1},$$
- where $s(t)$ is in feet and t is in seconds. Approximate its velocity at $t = 2$ by using Definition (3.3) with $h = 0.01, 0.001$, and 0.0001 .
- (c) 26 The position function s of an object moving on a coordinate line is given by
- $$s(t) = \frac{t - t^2 \sin t}{t^2 + 1},$$
- where $s(t)$ is in meters and t is in minutes.
- (a) Graph s for $0 \leq t \leq 10$.
- (b) Approximate the time intervals in which its velocity is positive.

3.2 DEFINITION OF DERIVATIVE

In the preceding section we used the limit

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad \text{or, equivalently,} \quad \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a},$$

in several different applications. This limit is the basis for one of the fundamental concepts of calculus, *the derivative*, defined next.

Definition (3.5)

The **derivative** of a function f is the function f' defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists.

The symbol f' in Definition (3.5) is read *f prime*. It is important to note that in determining $f'(x)$ we regard x as an arbitrary real number and consider the limit as h approaches zero. Once we have obtained $f'(x)$, we can find $f'(a)$ for a specific real number a by substituting a for x .

The statement $f'(x)$ exists means that the limit in Definition (3.5) exists. If $f'(x)$ exists, we say that **f is differentiable at x** , or that **f has a derivative at x** . If the limit does not exist, then f is not differentiable at x . The terminology *differentiate $f(x)$* or *find the derivative of $f(x)$* means to find $f'(x)$.

Occasionally we will find it convenient to use the following alternative form of Definition (3.5) to find $f'(a)$.